

ASAD HUSSAIN

EMBEDDED SYSTEMS · DIGITAL IC DESIGN (VERILOG/VLSI) · RF & SDR · PCB DESIGN · EV BATTERY SYSTEMS · APPLIED AI/ML

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PROFESSIONAL SUMMARY

Electrical Engineering candidate (B.E., NUST, 2027) seeking internship and graduate-engineer roles in embedded systems, digital/VLSI design, RF, or EV battery systems. Ships end-to-end systems that connect hardware, firmware and data: Altium PCB and analog design, Verilog and DSP, C/C++ firmware, and IoT/cloud back-ends with ML on top. Completed internships at five organizations (HIT, NCRA, NEPRA, FlyRank AI, CodeAlpha), built a production-grade fleet-tracking platform, and currently works as a Research Assistant on EV battery management (SOC/SOH estimation and cloud-connected BMS).

EDUCATION

National University of Sciences and Technology (NUST), Pakistan

2023 – 2027 (expected)

B.E. Electrical Engineering

Coursework: Digital System Design, Digital Communication Systems, DSP, Embedded Systems, Analog Electronics, Power Electronics, Control Systems, Instrumentation, Microprocessor Systems, OOP. **Honor:** Highest Achiever, SSC.

TECHNICAL EXPERTISE

Embedded & Hardware: STM32, ESP32, Arduino, PIC/AVR, Raspberry Pi; PCB design (Altium Designer, EasyEDA, Proteus); analog and power-supply design; bench debugging.

Digital IC & Signal Processing: Verilog, VHDL, VLSI design, DSP kits, FEC codec design, FFT/spectrogram analysis, MATLAB/Simulink.

RF & Communications: SDR (LimeSDR, RTL-SDR), GNU Radio, antenna design, digital communication systems, MQTT over cellular (GPRS/LTE).

EV & Battery Systems: BMS monitoring circuits, State of Charge (SOC) and State of Health (SOH) estimation, cloud-connected battery telemetry.

Programming & Backend: C/Embedded C, C++, Python, JavaScript, SQL, Flask, FastAPI, REST APIs, Docker, Git, Linux.

AI/ML: TensorFlow, PyTorch, Scikit-learn, LangChain/RAG, LLM fine-tuning, CI/CD and MLOps basics, n8n automation.

PROFESSIONAL & RESEARCH EXPERIENCE

Research Assistant, Electric Vehicle Battery Management Systems

Present

Supervised by Dr. Hassan Khalid, NUST

- Researching next-generation BMS technology for electric vehicles: architectures for cell monitoring, protection and balancing.
- Studying State of Charge (SOC) and State of Health (SOH) estimation and how both feed range prediction and battery-lifetime decisions.
- Investigating cloud support for BMS: streaming pack telemetry to the cloud for remote monitoring, analytics and model updates.
- Outcome:** designed a BMS monitoring circuit combining current and voltage sensing with an AI model that predicts battery energy consumption and estimates remaining vehicle range.

ON-SITE INTERNSHIPS

National Electric Power Regulatory Authority (NEPRA), Islamabad

Jul – Aug 2026

Technical Intern, Technical Department

- Mapped NEPRA's regulatory chain (license, tariff, sale) across generation, transmission, distribution and supply licenses, including all three generation tariff mechanisms.
- Outcome:** analyzed NGC transmission data for 233 grid-station transformers and found 116 (about half) operating above 80% of rated capacity; reviewed the State of the Industry Report covering 111,466 GWh of national consumption.
- Studied advanced metering infrastructure (PLC and cellular links, DLMS/COSEM) and power-transformer construction, ratings, losses and protection.

Research & Indigenous Development Centre (RDC), Heavy Industries Taxila(HIT)

Jun – Aug 2025

Engineering Intern, Full-Stack Systems

- Designed and tested a driver-logging device around the LilyGO T-SIM7670E (ESP32 + LTE), GPS, an RFID reader and a TP4056 LiPo charger; C/C++ state machines handle 1 Hz GPS capture, RFID driver authentication and buzzer feedback.
- Implemented MQTT (QoS 1) telemetry over cellular GPRS with about 50 ms server-side callback processing, and an asynchronous, multithreaded Flask + SQL Server back-end so analytics never block incoming streams.
- Outcome:** delivered a working fleet-management platform with ray-casting geofencing, harsh-driving detection, SD-card resilience and a Leaflet.js/Chart.js dashboard rendering each point in under 100 ms.

National Centre of Robotics and Automation (NCRA), NUST

Jul – Aug 2024

Research Intern, Robot Design & Development Lab (six weeks)

- Contributed to research and development in embedded systems, IoT and communication technologies; worked extensively with the ESP32 on hardware interfacing, programming, automation and system development.
- Outcome:** implemented Wi-Fi and Bluetooth communication, wireless command handling and device control; certified as consistent with the lab's standards for dedication and performance.

VIRTUAL INTERNSHIPS

FlyRank AI, Chicago, USA (Remote)

Aug 2026 – Present

Machine Learning Intern

- Built a content-opportunity scoring pipeline that flags pages with strong search visibility but low click-through rate, so content teams can prioritize refreshes.
- Worked through a staged ML workflow — data contracts, exploratory analysis, then baseline modeling — applied to real SEO and content datasets rather than a toy pipeline.
- **Outcome:** case-study write-ups translating model output into content-refresh recommendations now give stakeholders a direct line from model score to editorial action.

FlyRank AI, Chicago, USA (Remote)

Jul – Sep 2026

Backend AI Engineering Intern

- Completed practical assignments spanning API design, prompting, retrieval and evaluation, and attended the program's scheduled sessions, workshops and events throughout.
- Built an authentication-protected CRUD API on a connected database, containerized it with Docker, and automated a data-scraping workflow pipeline.
- **Outcome:** capstone project accepted — an LLM usage metering and billing service — and independently qualified in the AI Fluency track through 2 Anthropic Academy courses.

CodeAlpha (Virtual Internship Program)

Jun – Jul 2026

Machine Learning Intern

- **Delivered three ML projects:** a credit-scoring classifier (Logistic Regression, Decision Tree, Random Forest), a CNN for handwritten-character recognition on MNIST/EMNIST, and a disease-prediction classifier (SVM, Logistic Regression, Random Forest).
- **Outcome:** each project was benchmarked across model families on precision, recall, F1 and ROC-AUC, and all three now anchor the Machine Learning section of my project portfolio.

FEATURED PROJECTS

Final Year Project: Lite Distributed Semantic Communication System with FEC (*MATLAB, Verilog, VLSI, DSP kit*) — Modeled a semantic communication system with a forward-error-correction encoder/decoder in MATLAB, then implemented and verified a synthesizable Verilog pipeline on a DSP development kit for robust message reconstruction under channel noise.

Multi-Stage Audio Power Amplifier — Designed and simulated a multi-stage audio amplifier (CS pre-amp, CE gain stage, and diode-biased Class AB power stage) achieving an overall voltage gain of 694 V/V across the 20 Hz – 20 kHz audio spectrum.

AMI-Based IoT Smart Energy Meter (*ESP32, Proteus, full-stack dashboard*) — Metering node with a web dashboard taking live energy consumption from ESP32 Lilygo sensor readings to a utility-facing view.

Analog Circuit Design (*Op-amp/RC, transformer, regulation*) — 2nd-order Butterworth low-pass filter (over 10 dB attenuation at 11.25 kHz, flat 20 dB passband to 900 Hz) and a benchtop DC power supply; both simulated, built and bench-validated.

PCB Design (*Altium Designer, EasyEDA*) — ESP32 USB communication interface board (Altium); two-board ESP32 system with a sensor board (MAX30102, MPU6050, MLX90614) and GSM telemetry (EasyEDA).

Academic Projects — EEG cleaning MATLAB GUI (Parkinson's data), 5-band graphic equalizer, PID DC-motor controller, 4-bit ALU, FM receiver (TDA7000), multi-stage audio amplifier.

TRAINING & CERTIFICATIONS

atomCamp AI Bootcamp: Pakistan's premier AI training institute partnered with Meta & Microsoft

- **Deep learning and NLP/LLMs:** neural-network foundations and how large language models are built and used.
- **Generative AI and RAG:** built and deployed a knowledge-bot agent (LangChain, FAISS, OpenAI APIs) with a Streamlit interface.
- **MLOps:** containerized with Docker and served through FastAPI; CI/CD, deployment and monitoring basics; n8n workflow automation.
- **CS50x, Harvard University (2026):** core computer science (algorithms, data structures, C++, Python, SQL, web), proven through ten problem sets and a final project.
- **PCB Basic Design, Altium Education (2026):** schematic capture and PCB layout workflow in Altium Designer.
- **Effective Leadership, Deloitte WorldClass (Aug 2026, scored 90%):** leadership fundamentals for guiding and motivating teams.
- **Data Analysis with Python, IBM (Apr 2024):** data wrangling, analysis and model development with Python (pandas, NumPy).
- **Digital Product Management, Alison (Apr 2024):** digital product lifecycle and management fundamentals.
- **MathWorks MATLAB:** MATLAB fundamentals for numerical computing and analysis.

COMPETITIONS, LEADERSHIP & ACTIVITIES

- **Voltfest 2026 (NUST Robotics Society):** led the computer-vision/object-detection initiative as Technical Lead and demonstrated a real-time YOLOv8 detection dashboard.
- **SEMPEC – Project Display Competition (organized by BRISC with BIOMISA Lab):** participant; showcased a semester project (PID Motor Controller using Arduino Uno and LM298) to peers and visitors in a competition.
- **NMUN 2024:** delegate in Model UN committee sessions, building research, negotiation and public-speaking skills.
- **Technical Lead, NUST Robotics Society; member, NUST Engineering Society:** organized PCB-design bootcamps for peers and robotics/electronics workshops for high-school students (circuit design, PCB layout, Arduino/ESP32).

Professional Membership: Member, [Institute of Electrical and Electronics Engineers \(IEEE\)](#)

Languages: Urdu (native), English (professional working proficiency), French (Elementary)